

Simulation Final Exam Questions and Answers

Azarbaijan Shahid Madani University - Second Semester of
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Question 1

Using Monte Carlo computational principles, find the area of the hatched region in the figure below:

- $y = -x$
- $y = 2x$
- $y = 4$

Answer:

First, we find the intersection points of the lines to determine the bounding rectangle:

- Intersection of $y = -x$ and $y = 4$: $x = -4 \Rightarrow (-4, 4)$
- Intersection of $y = 2x$ and $y = 4$: $x = 2 \Rightarrow (2, 4)$
- Intersection of $y = -x$ and $y = 2x$: $x = 0 \Rightarrow (0, 0)$

Thus, the range of the bounding rectangle will be $-4 \leq x \leq 2$ and $0 \leq y \leq 4$. The area of this rectangle is equal to:

$$S_{\text{rect}} = (2 - (-4)) \times (4 - 0) = 6 \times 4 = 24$$

Monte Carlo Algorithm: We must generate N random points with a uniform distribution inside the bounding rectangle and check how many of these points (M) fall within the hatched region. The condition for falling within the hatched region is as follows:

$$y_i \leq 4 \quad \text{and} \quad y_i \geq -x_i \quad \text{and} \quad y_i \geq 2x_i$$

The area of the region (A) is approximated via the following formula:

$$A = \frac{M}{N} \times 24$$

For example, if we generate $N = 100$ points and assume $M = 50$ points fall within the region, the approximate area according to the formula will be 12. According to the geometry of the figure, the actual area is also exactly 12.

Question 2

We intend to generate 10 random numbers whose mean is approximately 12 and whose standard deviation is approximately 2. How should we generate these numbers? (Construct at least one of them). How can we simulate the arrival of bank customers using these numbers?

Answer:

To generate random numbers with a mean of 12 and a standard deviation of 2, we use the standard normal distribution relation. The random value is obtained from the following formula:

$$X = \mu + Z\sigma \Rightarrow X = 12 + 2Z$$

Next, for 10 numbers, we randomly select values of Z from the standard normal distribution and substitute them into the formula. As an example, if the generated random number for Z is equal to 2:

$$X = 12 + 2(2) = 16$$

We repeat this process 9 more times to generate 10 random numbers with a mean of 12 and a standard deviation of 2.

Simulating Bank Customer Arrivals: For this purpose, the normal distribution can be used. In such a way that each of the numbers obtained in the previous step is considered as the number of customers in a time interval (e.g., a one or two-hour interval) to analyze and forecast. For example, suppose that in the first 2 hours 16 people and in the next two hours 14 people enter the bank. These data can be used to analyze queue length, waiting time, and determine the optimal number of servers.

Question 3

For cities A, B, and C according to the figure below, a traveler can move with the given probabilities. What should the three probabilities p_1 , p_2 , and p_3 be? Construct the Markov transition matrix. If the traveler is in city A at time $t = 0$, with what probability will they still be in city A at time $t = 2$? At time $t = 50$, which city will the traveler be in? Why?

Answer:

Given that the sum of outgoing probabilities from each node (city) must be equal to 1:

- From city A there is only one outgoing path (to B), so $p_1 = 1$.
- From city B the return path to A has a probability of 0.3, so the probability of going to C is equal to $p_2 = 1 - 0.3 = 0.7$.

- City C has no outgoing paths to other cities (absorbing node), so $p_3 = 1$.

Markov Transition Matrix (P):

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0.3 & 0 & 0.7 \\ 0 & 0 & 1 \end{bmatrix}$$

Since the traveler is initially in city A, the initial state vector is as follows:

$$V_0 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

State at $t = 1$:

$$V_1 = V_0 \times P = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0.3 & 0 & 0.7 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$$

State at $t = 2$:

$$V_2 = V_1 \times P = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0.3 & 0 & 0.7 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.3 & 0 & 0.7 \end{bmatrix}$$

Therefore, the probability that the traveler is in city A at time $t = 2$ is equal to the first element, which is **0.3**.

State at $t = 50$: City C is an absorbing state. According to the paths, if the traveler starts from A they will definitely go to B, and when in B, with a much higher probability (70 percent), they go to C. If they go to C, they can no longer go anywhere else and remain in that same city. Due to the high number of steps ($t = 50$), the traveler will almost certainly be in city C (the probability of remaining in the cycle of A and B becomes negligible).

Question 4

The movement direction of cars in lanes 1 and 2 is specified as shown in the figure. In lane 1, an average of 10 cars enter per minute. In lane 2, an average of 4 cars enter per minute. Lane 1 has the right of way, and the three-way intersection lacks traffic lights. Each driver in lane 2 has a 30 percent probability of an accident. If 100 cars enter from lane 1, approximately how many car accidents will occur at the intersection? Simulate this system as a computational model.

Answer:

Mathematical Calculations: The arrival rate of cars in the first lane is $\lambda_1 = 10$ cars per minute. The time required for 100 cars to pass through lane 1 is:

$$T = \frac{100}{10} = 10 \text{ minutes}$$

The expected number of cars in the second lane during these 10 minutes with a rate of $\lambda_2 = 4$ cars per minute is equal to:

$$N_2 = \lambda_2 \times T = 4 \times 10 = 40 \text{ cars}$$

The probability of an accident for each car in the second lane is $p = 0.3$. This process follows a binomial distribution $X \sim \text{Binom}(40, 0.3)$. The expected number of accidents is equal to:

$$E(X) = n \times p = 40 \times 0.3 = 12 \text{ accidents}$$

Simulation Model: The simulation algorithm is proposed as follows:

1. Generate the arrivals of cars in lane 1 with a Poisson rate of 10 cars per minute.
2. Continue the simulation until 100 cars pass through lane 1.
3. Generate the arrivals of cars in lane 2 with a Poisson rate of 4 cars per minute within the determined time interval.
4. For each car in lane 2, we generate a uniform random number U in the interval $(0, 1)$.
5. If $U \leq 0.3$, an accident is recorded.
6. Finally, the total accidents are reported, the average of which over multiple runs will be approximately equal to 12.